

Manual

Escalation Management (Basis/Framework) SIS-ESK 4.0_{pe}

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1 Escalation Management - Overview

Purpose

HYDRA escalation management provides a framework of functions that can be used to forward events that occur or were recorded in HYDRA to individual users or user groups in real time. The function of escalation management is to take active steps to ensure users are notified.

After notification, escalation management monitors times until acknowledgment by the recipients and until escalation is concluded. Escalations can be forwarded to other users or user groups during processing.

Implementation considerations

You use escalation management function if you request an active and early notification of specific events in the production environment in order to prevent downtimes and increase efficiency and productivity by early response.

You use the escalation management function if you need to monitor the response times to escalation of specific departments or persons.



The usage of multilingual database content (MDB) is not supported by escalation management.

Integration

Escalation management forms the framework for other HYDRA components in order to be able to transfer and track events triggered there.

For notifying individuals, the escalation management accesses both the [user administration](#) and [HR master data](#) registered in the system.

If the Recording and Maintenance of Labor Times component (PZE-EPP) is used, the escalation management function allows for identifying the addressee subject to the currently recorded attendance.

Notifications can be sent as e-mails by integrating the local mail server into the system.

Features

Escalation management features the following functions:

- Addressee definition:
 - Allocation of individual persons as addressees for messages

- Summary of several individuals into a function group as the recipient of a message
- Notification of user:
 - Notification of users of the MES Operation Cockpit in the MOC
 - Notification of recipients by e-mail
- Integration with HR:
 - Verification of recipients' attendance by comparison with In/Out statuses of individuals as recorded by the HYDRA PZE.
 - Definition of substitutes if the employee to be contacted first is not available (information from above-described comparison)
- Dispatch of messages on basis of flexible condition configuration
- Message management
 - Signing function for messages / forwarding on MOC
 - Possibility for repeated forwarding of messages to another user
 - Evaluation for traceability of escalation and monitoring response times
- Provision of escalation:
 - Triggering of escalation if not valid recipient could be identified
 - Triggering of escalation for unprocessed escalation

Component structure

The core of escalation management is the escalation manager. This is a continuously active process (hyeskmgr.out/hyeskmgr.out) which receives events generated by individual applications and verifies whether escalations have been configured for these events. If this is the case, the escalation manager generates messages to users or groups of users for recorded communication technologies in accordance with the configuration.

The events themselves are generated by the individual HYDRA applications. The causes of events may be error or exceptional situations on the one hand (e.g. if a disk is full), or the result of "normal" recording and processing of HYDRA on the other hand (e.g. if a new machine status is recorded on the terminal).

In such cases, the applications generate an event. This is identified by an unambiguous designation in the entire system. In addition, the applications provide data to the escalation manager. The type and scope of the transferred data depend on the relevant event. In many cases, such data may be used as conditions in the configuration and integrated into the messages to be sent.

Not all of the applications are capable of generating a message in each error case or exceptional situation (e.g. a terminal is not able to report that it is offline). For this reason, the escalation manager has another component - the escalation agent. The escalation agent periodically verifies the current status for values allocated to periodical monitoring. If violations occur during verification, the controlling agent sends the relevant event to the escalation manager. Once an hour, the escalation manager updates its internal list of configured events using the periodical events recorded in the data base.

To perform error analysis, you can use the "hyeskmgrpro" file containing "TRUE;D+t" to activate a log (./err/hyeskmgr.err.pro) at runtime in which, among other things, the communication is recorded while sending the message (e.g., e-mail).

Available escalations

Escalation management as described in this documentation is an integral part of the MES weaver. Escalations are triggered by individual applications to be allocated to the individual application packages. In addition, available escalations depend on a release. For this reasons, please find a schedule of the escalations available for the relevant release and module in the relevant documents.

Event	Description	Identifiers	Description	Please note:
ESK.NO_PERSON_AVAILABLE	No person available	ESK.ESKID	Event for which no person could be identified	The event is triggered if no active person could be identified within a function group when checked for attendance on the console and/or the PZE.
		ESK.ESKBEZ	Event designation	
		ESK.RCV:FKT	Recipient function group	
ESK.UNPROCESSED_ESCALATION	Unprocessed escalation	ESK.ESKVERWEIS	Event ID Unambiguous event ID displayed in the escalation management; it is composed of the event and the clear reference.	The event is triggered if an escalation has occurred but remains unprocessed.
		ESK.VERWEIS	Reference Clear referent to event	
		ESK.ESKID	Event	
		ESK.ESKBEZ	Event description	
		ESK.MSGSUBJ	Message subject	
		ESK.MSGTXT	Message text	
		ESK.DATB	Date of the triggering	
		ESK.ZEIB	Time of the triggering	
		ESK.DAUER	Duration (in seconds)	
		ESK.RCV:ART	Type of recipient	
		ESK.RCV:BEARB	Recipient user name	

Event	Description	Identifiers	Description	Please note:
		ESK.RCV:PNR	Recipient staff number	
		ESK.RCV:FKT	Recipient function group (if available)	
		PNR.PVORNAME	Recipient first name	
		PNR.PNAME	Recipient (last) name	
		ESK.RCV:BEZ	Recipient	
HYD.LOCKED_RECORD	Locked record data	LOCK.APPLICATION	Locked application!	The event is triggered if a locking record blocks processing in the table hyd_lock. (e.g. if feedback is stopped)
		LOCK.USER	User who triggered the locking	
		LOCK.BEARB	Employee who triggered the locking	
		LOCK.MODUL	HYDRA module which triggered the locking	
		LOCK.KEY:1 LOCK.KEY:5	Locked values	
		LOCK.DAT	Date of locking	
		LOCK.ZEI	Time of locking	
		LOCK.DURATION	Duration of locking	

2 Escalation Configuration

Overview

Menu	Master data → Escalation management → Escalation configuration
Transaction code	esccfg
Function authorization	esccfg

Purpose

Use this function to specify rules for the conversion of events into escalations (events provided by the applications). In the escalation configuration, you define the events that are converted to escalations, the recipients of specified escalations and how the escalation is technically sent.

Integration

To convert events into escalations, the escalation management uses the data stored in the escalation configuration.

Requirements

You have defined [Function groups](#) in the system if you:

- want to send an escalation to several recipients at the same time.
- want to send an escalation to specific persons of a group subject to defined factors, e.g. attendance.

You have linked HYDRA users of the [User administration](#) to persons of the [HR master data](#) to be able to send messages to these users/persons.

Field descriptions

Tab Event

The *Event* tab includes general information on an escalation. You can maintain the following information:

Event

This event triggers an escalation. Applications generate the event if specific situations occur.

Using the search function, you can select an event from the list of available events. You cannot enter events that are not available/included in the list.

Designation

You can enter a description of the event in this field. The system automatically suggests the description stored with the event when you select an event. You can overwrite this suggestion.

The name entered in this field is later used in the overview dialogs of current escalations. To easily identify the event, it is best to store a description that refers to the event and to the formula.

Recipient type

You can define the type of the recipient of a message in this field. The following types of recipients are available:

Function	You can select a function group that is notified.
Person	You can enter a personnel number for this recipient type.
Supervisor	The notification is sent to the supervisor of the person included in the escalation data. The acronym "PNR" identifies the person.
Automatic	The notification is sent to the person included in the escalation data. The acronym "PNR" identifies the person.
Workflow management	The escalation triggers a specified workflow.

Recipient

Depending on the selected type of recipient, you can enter the recipient here.

If you have selected "Person" as recipient type, you can use the search function to select the person from the HR master data.

If you have selected "Function" as recipient type, you can use the search function to select the recipient from the stored function groups.

Priority

You can use the priority in the message overview to sort the messages. The priority does not have any other function.

Farbe

Currently not used.

Active

This option specifies if the event configuration is enabled or not.

Cyclic monitoring

If the selected event is cyclically monitored, you must maintain the monitoring cycle here. The cycle is defined in days, hours and minutes. The default values are preallocated when you generate the monitoring. You can overwrite these values.

The system provides the event within the time specified as cycle. The conditions that might be stored for the escalation specify, if an event triggers an escalation. Example:

You have specified a cycle of 15 minutes for the cyclic escalation ABC.XYZ. The system therefore identifies the required data in the system every 15 minutes and provides an event. If the conditions stored in the escalation are fulfilled, the system converts the event into an escalation.



The escalation agent updates the internal list of configured events with the cyclic events stored in the database once an hour.

Automatic: after reading the message

Currently not used.

Automatic: after closing the event

Currently not used.

Automatic: after sending the message

If the event includes an automatic mode, you can view, enable and if required reconfigure the automatic here – depending on the definition of the event. You can define an automatic function that is performed when the message is sent (in case of an escalation).

The following automatic functions are available (depending on the event):

Function Close:

This function closes the event. You can configure that for example the escalation is automatically closed once the message has been sent.

If the function "Close after sending the message" is enabled, the message is sent and the event is closed with the comment "-".

Find below an example for the generation of a template:

You want to configure an escalation for the event TNR.OFFLINE for the terminals 11-14 that often cause problems. You want to forward this event to the system administration for a quick troubleshooting. Make the following entries in the *General* tab:

Field	Entry
Event	TNR.OFFLINE
Description	Terminal 11 - 14 OFFLINE
Recipient type	Function group
Recipient	Maintenance
Cyclic monitoring	0 hours, 00:30 minutes (the terminal status is checked every 30

minutes).

Workflow management

Workflow

Select the workflow that is triggered in case of this event.

User

HYDRA user that is transferred with the triggered workflow as user. Depending on the workflow, the defined user is also used in the workflow.

Group

Workflow group that is transferred with the triggered workflow as group. Depending on the workflow, the defined group is also used in the workflow.

Function

Workflow functions that are transferred with the triggered workflow as function. Depending on the workflow, the defined function is also used in the workflow.

Tab Options

Cyclic monitoring

This option defines if the event is generated via cyclic monitoring.

ESK mode

Type of escalation processing (currently only "F").

Automatic: after reading the message

Currently not used.

Automatic: after closing the event

Currently not used.

Automatic: after sending the message

See tab *Event*.

Tab Message

In the *Message* tab, you can maintain templates for the messages to be sent. Here, you enter the required message text and, if the selected type of notification is e-mail, also the subject.

Each event provides additional detail data. The detail data that an event provides is defined in the system. When you generate the template for a message, you can use placeholders for the provided detail data. If an event triggers an escalation, the placeholders in the template are replaced by the current values of the event and the complete current message text is generated.

You can maintain the following fields:

Subject

The text entered here is used as subject if the notification type e-mail has been selected. Also in the subject line, you can use placeholders that are replaced by the current values later on (see also message).

The system automatically uses the event name stored in the system as subject when the event is created. You can overwrite it.

Text

In the field *Text*, you edit the template for the actual message. You can store up to 320 characters for the message text. Note:

The maximum length depends on the message type. The following lengths are currently available:

E-mail: 320 characters

Text message (SMS): 320 characters (2 x 160 characters)

Pager: only the subject is transferred

The defined template might increase in size when the placeholders are replaced. When generating the message text, it might happen that the resulting text exceeds 320 characters. In this case, the placeholders cannot be replaced. With e-mail as communication type, the complete message text can be transferred.

In the template, you can store placeholders that are replaced by current values later on. The list of available placeholders shows the available placeholders.

Available placeholders

You can include placeholders in the message text, if you enter the identifier listed in the *Available placeholders*. Put a "%" before and after the identifier. Double-click the list of available placeholders to insert the selected placeholder at the end of the text template.

The message "Terminal 11 is offline" is generated from the template "Terminal %TNR.TNR% is offline" if this event occurs on terminal 11.

In the following, the example of the previous section is refined. In the *General* tab, you have already edited data for the event TNR.OFFLINE for the terminals 11-14. You want to notify the system administrators via e-mail and via text message (SMS). You can use a maximum of 2x160 characters (max. message length). The list of available placeholders displays the following detail data that is available for the event TNR.OFFLINE:

TNR.BEZK → Terminal location

TNR.BEZL → Terminal designation/name

TNR.TNR → Terminal number

TNR.ZYKL:I → Actual cycle of status messages

TNR.ZYKL:S → Target cycle of status messages

System variables

In addition, the placeholders DAT (system date) and ZEI (system time) are available. They include date and time of message receipt in the server and can be formatted as follows:

%ZEI time hh:mm:ss% or

%ZEI time hh:mm%

or

%DAT date dd.mm.yyyy% or

%DAT date mm/dd/yyyy% or

%DAT date yyyy-mm-dd%

The placeholder WTG (weekday) includes the weekday of the system date. The placeholder is replaced by 3 characters describing the English weekdays: MON, TUE, WED, THU, FRI, SAT and SUN.

The placeholder BEM (comment on the escalation) is available for specific escalations. The escalation descriptions specify whether and how this parameter is assigned

The system provides the placeholders MSGPRIO, MSGCLASS and MSGRCV for specific escalations. The system collects the values priority/urgency (MSGPRIO), information class/importance (MSGCLASS) and recipient/addressee (MSGRCV) for a posting (e.g. reset maintenance in the terminal). These values are then transferred in the escalation (e.g. escalation RES.MAINTENANCE_RESET). You can refer to these values in the condition and in the message.

- The placeholder MSGPRIO includes the priority/urgency of a posting and can have the following values: empty (not specified), 1 (highest priority), 2 (high priority), 3 (normal priority), 4 (low priority) and 5 (lowest priority). If the message is sent via e-mail, the priority is taken over as mail priority.
- The placeholder MSGCLASS can include the following values: empty (not specified), I (information), W (warning) or E (error).
- The placeholder MSGRCV includes the group of addressees (not function group!), e.g. "plant manager".

Make the following entries in the *Message* tab:

Field	Entry
Subject	Terminal %TNR.TNR% OFFLINE

Text Terminal %TNR.TNR% at location %TNR.BEZK% has been offline at least since %DAT date dd.mm.yyyy%, %ZEI time hh:mm:ss%. Please check network availability.

Tab Notification

The *Notification* tab defines how the person or function group stored in the *Event* tab is notified. You can configure different notification combinations using logic combinations.

The notification types defined in this tab apply to all persons of a function group. Here, the system does not check if the required communication data is maintained in the HR master data for individual persons or for the member of a function group and if the data is maintained in the basic settings.

Currently the following notification types are available: Notification to the client (always possible) and notification via e-mail, SMS/text message and pager (if licensed). You can always configure these notification types – independent of licensing. The notification itself is only performed with the respective license.

You edit the notification type as follows:

Operator

The operator specifies how the notification types are combined with each other. You can use the following operators:

"TO"

The message is sent using the message type stored below.

"AND"

The message is additionally sent using this communication type.

"OR"

For the notification type of the higher level, the system checks if the notification can be realized successfully (it is checked if all data required for the notification is available). If this is not the case, the notification is sent using the notification type of this level.

You can therefore use different individual communication types within a function group.

The notification type is maintained top down with descending priority. For technical reasons, the system uses different methods to check the success of the individual notification types:

- Notification on the client

You can immediately check if the notification has been successful. If the person is currently logged on, the person is notified and the message is immediately displayed in the current message window.

- Notification via e-mail

Promptly after generation, the e-mail is transferred to the local SMTP server (mail server). If the transfer could be performed, the notification is identified as successfully transmitted.

If the e-mail delivery fails (e.g. due to a wrong e-mail address or because the server is not available), the escalation management does not correct the delivery success (e.g. by sending another message via a different communication channel).

- Notification via SMS/text message

Promptly after generation, the SMS/text message is sent using the licensed transmission process. The message is identified as successfully transferred if all data required for the transmission (phone number, directories, etc.) is maintained.

If the text message delivery fails (e.g. due to a wrong phone number or because the server/gateway is not available), the escalation management does not correct the delivery success (e.g. by sending another message via a different communication channel).

- Notification via (PZE) terminal

With this notification type it is configured that the message is displayed only once on the PZE terminal and it is removed after a maximum of 4 weeks. However, the message can be displayed several times during the period of the cyclic loading of authorizations. The message is displayed with the following actions on the PZE terminal: In, out, break, info, message and absence reason.

Messages can only be displayed on PZE terminals of type CT-36x, CT-37x and CT-38x. For more information, refer to the document "[Configuration Absence Workflow](#)".

- Notification via ERP system

The SAP Alert Management provides functionalities within the ECC to actively inform users of events that have occurred. If you additionally use the SAP Enterprise Portal, you can view and edit these alerts according to the requirements of the user.

If the escalation management is integrated into the SAP Alert Management and the SAP Enterprise Portal, you can transfer the generated escalations into the SAP Alert Management. The escalations are then available for the users of the SAP Enterprise Portal. This option requires the license ESK-SAPALM. For further information, please refer to the documentation ESK-SAPALM.pdf.

Send e-mail in cc to

Here, you can store an e-mail address that receives a copy of the sent e-mail.

The system only sends this copy, if you have configured the notification via e-mail. If you have exclusively configured other types of notification, an e-mail copy is not sent. The stored cc e-mail address is not used.

We use the example of the previous sections and configure below how the system administration is notified. The system administration does not work all the time, but the console agent is installed in the workplace PCs. But as the employees are not always present at their workplace, a second notification is sent via SMS/text message.

Make the following entries in the *Notification* tab:

Field	Entry	
	Operator	Notification type
Level 1	TO	Console
Level 2	AND	SMS/text message

Tab Condition

In the *Condition* tab, you define the conditions that trigger an escalation for an event. With each event, the escalation manager checks the conditions that are stored here. Only if the conditions are fulfilled and the result of the check is true, the escalation is generated or triggered.

You define the conditions using formulas and logical expressions. You link the expressions using comparison operators. You can combine and nest expressions using parenthesis .

In the formulas, you can include identifiers that can be used for conditions in the system. Find below the available identifiers for the conditions that the application transfers with the event.

You can use the following operators and expressions to formulate conditions:

"and"

Use the expression "and" to formulate AND constructions. An AND construction has the following structure:

< condition 1> and <condition 2 >

This condition is only true, if BOTH conditions are fulfilled.

This means: The condition "it rains and storms" is only true if it rains and storms at the same time. If it only rains and does not storm (or vice versa), the condition is not fulfilled.

"or"

Use the expression "or" to formulate OR constructions. An OR construction has the following structure:

< condition 1> or <condition 2 >

This condition is only true, if at least one of the two conditions is fulfilled.

The expression "it rains OR storms" is true if it only rains or only storms. If both expressions are true – it rains AND storms – the condition is also true.

"like"

Use the operator "like" to compare characters. You can use the operator as follows:

< value 1> like < example value 1 >.

For the example value, you can replace leading or trailing characters by additional placeholders:

"*" 0 –n any characters

"?" exactly 1 character (any)

The comparison "name like SCHMID*" is true for SCHMID and for SCHMIDT and also for SCHMIDLER. But it is not true for SCHMITT. If the comparison were "name like SCHMI*", it would be true for all names beginning with SCHMI, i.e. SCHMID, SCHMIDT, SCHMIDLER and SCHMITT.

"=="

If you use the operator "==" the condition must be fulfilled 100 %. You can use the operator to compare characters and to compare numbers. Use it as follows:

< value 1 > == <example value 1>

For the example mentioned above, the condition "name == "SCHMIDT"" is only true for SCHMIDT.

"!="

If you use the operator "!=" (not equal), the condition must not be fulfilled. You can use the operator to compare characters and to compare numbers. Use it as follows:

< value 1 > != <example value 1>

For the example mentioned above, the condition "name != "SCHMIDT"" is not true for SCHMIDT, but for all other names.

"<", ">", "<=", ">="

You can use the operators "<", ">","<=" , ">=" to compare numbers (and characters). Use them as follows:

< value > ">" <value 2>

The operators have the following meaning:

"<" less than...

">" greater than...

"<=" less than or equal to....

">=" greater than or equal to...

Make the following settings to store a condition:

Condition

Maintain the logical condition in the text field of the condition. You can use the above mentioned expressions, operators and combine them using parenthesis. You may only use identifiers that are provided by the event. The section "Variables" displays the available identifiers. If you use the identifiers in the condition, you must EXACTLY take over the syntax presented in the variables section.

A link at the end of this section includes an overview of functions, operators and constants that can be used in formulas and conditions.

If you use decimal numbers in formulas, use the dot "." as decimal separator.

Variables

This field shows the identifiers that are available for the respective event. In this field, you can also enter values for each identifier. You can store values that are expected in the application. Using these values, you can then immediately check if the formula is correct.

To check the result of the formula, the collected value for each identifier is entered in the formula and the formula is calculated. The formula returns the value "TRUE" or "FALSE" as a result. If the return value is "FALSE", you must either check the formula or the values. You must use ">"<" to begin and end strings, e.g. descriptions or words.

If you double-click a possible variable, this variable is inserted at the end of the formula.

System variables

System variables are provided in the same way as the variables.

In the previous chapters, it was discussed how to configure the notification of the system administration, if terminal 11-14 are offline. To complete this example, the formula is now stored.

The system administration should be notified, if at least one of the terminals 11 to 14 is affected. The event TNR.OFFLINE provides the identifier TNR.TNR (terminal number). The basic condition must be:

TNR.TNR >= 11

Here, the system would always trigger escalations if the terminal number were greater than or equal to 11. But you only want to generate messages, if the terminals 11-14 are affected. The condition is therefore extended.

TNR.TNR >= 11 and TNR.TNR <= 14

The condition is only true, if the terminal number is greater than or equal to 11 on the one side and on the other side it is less than or equal to 14.



Find a complete list of the supported operators [here](#).

Tab Condition – Examples for conditions

If you define conditions, you must make a difference between the data types CHAR (for characters) and INTEGER for numeric values or periods (of time).

If you compare CHAR data types, you must always enclose these CHAR data types by double quotes. You can compare numeric values without double quotes.

To find out if an acronym/field is numeric or if it is a string, you can check how the field is maintained in the system. If you can enter letters into the respective field, you must enclose the value by double quotes.

Example:	Condition
Machine status change to status 10 (MST.MALFUNCTION_OCCURRED) at a specific machine (alphanumeric) should trigger an escalation.	MNR.MNR == "WP1" and MST.MST == 10
Machine status change to status 10 (MST.MALFUNCTION_OCCURRED) at a specific machine (numeric) should trigger an escalation.	MNR.MNR == "123" and MST.MST == 10
A machine status 6 that exists longer than 10 minutes (MST.MALFUNCTION_CONTINUE) should trigger an escalation.	MST.MST == 6 and MST.DAUER >= 600
A machine status 6 that exists longer than 10 minutes at a specific machine (MST.MALFUNCTION_CONTINUE) should trigger an escalation, but only at a specific machine	MST.MST == 6 and MST.DAUER >= 600 and MNR.MNR = "1234"

(numeric).	
A machine status 6 that exists longer than 10 minutes at a specific machine (MST.MALFUNCTION_CONTINUE) should trigger an escalation, but only at a specific machine (alphanumeric).	MST.MST == 6 and MST.DAUER >= 600 and MNR.MNR = "WP1"
You want to check if less than 90 % of the target quantity (ANR.SGR:GUTP) of an operation is posted (ANR.EGR:GUT is the actual quantity)	ANR.EGR:GUT < (0.9*ANR.SGR:GUTP)

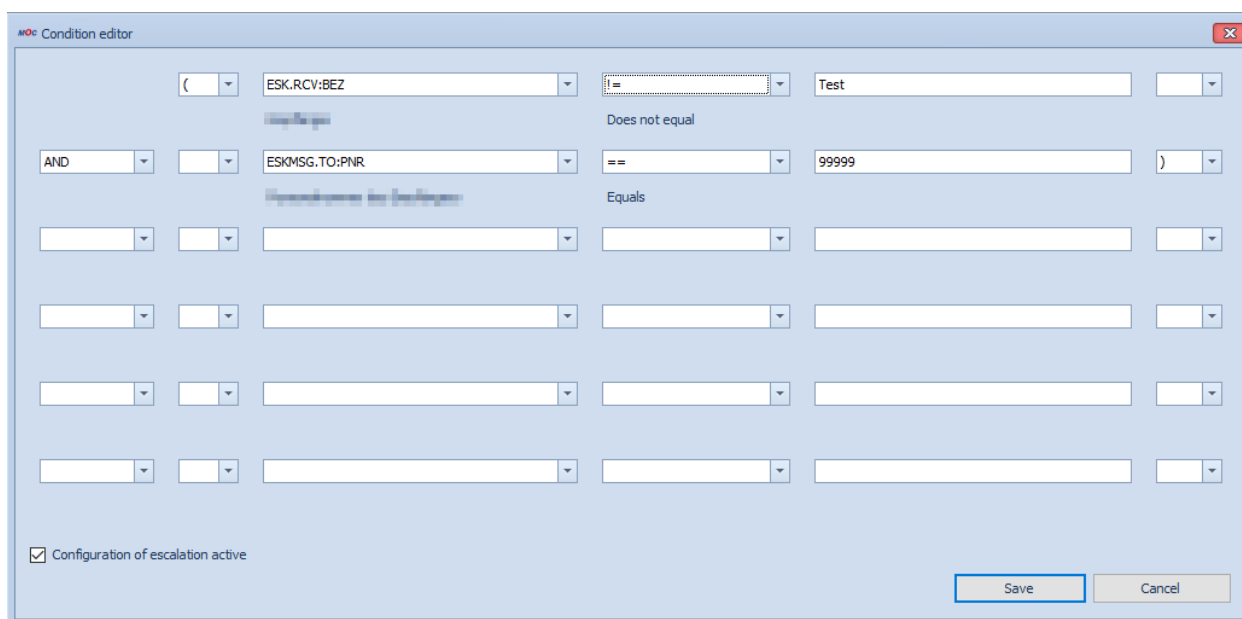
Condition editor

The conditions in the data records of the escalation configuration can become very complex. The *Condition editor* helps to edit the conditions and supports the user in the configuration of valid conditions.

The condition editor loads the condition of the data record currently selected. When the condition is edited and the condition editor is closed, then the condition is updated in the current data record.

If formula parsing is not possible, an error is displayed. If a formula stored in the system cannot be read when the condition editor is opened, an error is displayed.

Click the button *Condition editor* in the toolbar to call the condition editor. If a data record is selected, this data record is displayed in detail in the condition editor.



You can use the checkbox in the *Condition editor* to specify whether the escalation configuration is active or not.

Click the button *Save* to write the data to the current data record.



The condition editor does not know the data type of the variable and assumes that it is data type "string". If you use numeric variables with comparison operators with "greater" or "less", you must edit the created condition manually after using the editor. Delete the quotation marks. The comparisons will then work properly.

Tab *Last change*

Modified by

Last person who edited the escalation

Modified on

Date and time of the most recent modification

3 Function Groups

Overview

HYDRA menu	Master data → Escalation management → Function groups
FEDRA menu	Detailed scheduling → Master data → Escalation history
Transaction code	escfg
Function authorization	escfg

Purpose

You use this function to create or modify function groups in the system.

Integration

Use function groups in the escalation management to

- notify several recipients at a time;
- identify specific recipients within a group (e.g. a group of maintenance engineers) that meet special requirements, e.g. the recipient is present.

Requirements

You have defined the recipients as person in the [HR master data](#) and:

- if you want a notification via e-mail, you have created the e-mail address of the person;
- If you want a notification in the MOC, you have linked the user in the [User administration](#) to a person of the HR master.

Field descriptions

Function

The function is the name of the function group. If function groups already exist, you can select the group via the detail selection.

Priority

The priority defines the notification order within a function group. You can assign the same priority to several persons within a function group.

The highest priority is "1", the lowest priority is "999". The priority helps the system to deliver the messages.

Personnel number

The personnel number identifies the person that should be notified.

** Check console**

If you have checked the option, the system will verify if the person is logged on to the client. If the person is not logged on, the system verifies the person with the next priority level. You use this setting, if you do not use the "time and attendance" system. In case of a negative result, i.e. the person is not logged on to the client/console, the message is not delivered in an alternative form (text message, e-mail...).

Example:

Three maintenance engineers work in three different shifts. If you assign within a group the same priority to all three of them, only the engineer actually logged in to the console receives the message.

If you want to use the function "Check console", you must have entered the personnel number in the User administration.

** Check attendance**

If you have checked the option "Check attendance", the system will verify, if the respective employee is logged on to the PZE system "time and attendance" (only in connection with PZE-BP).

The escalation management will suppose that the person is present, if the following PZE events occur: In, break, entrance before clocking in (with access control system ZKS).

There is an AND-link between the settings "Check console" and "Check attendance". If both options are checked, the person only receives messages when he or she is present AND logged in to the console.

That is the reason why we recommend to check the option "console" OR the option "attendance".



The system cannot send a message to a function group, if no one of this group is logged in to the console or is logged in to the PZE system "time and attendance".

In this case the escalation management will generate the event "NO_PERSON_AVAILABLE". You should assign a dispatcher group to this event or you should search for this event on a daily basis in order to find cases left aside.

4 Current Escalations

Overview

HYDRA menu	Information management → Messages → Current escalations
FEDRA menu	Detailed Scheduling → Current → Current escalations
Transaction code	escov
Function authorization	escov escov.forward – Forward escalations escov.close – Close escalation escov.disp – Show messages of all users (Dispatcher rights)

Purpose

Use this function if you want to:

- get an overview of your escalations/messages (or the ones of others).
- to confirm receipt of a message, forward or close a message.
- to get an overview of completed messages.

Integration

The evaluations/reports show escalations of all components of the system.

Requirements

- You have activated the *Escalation Management* module.
- You have configured events and assigned these configurations to recipients.

Selection criteria

In general, the overview only shows escalations that:

- are intended for the logged in user.
- have the status "open" or "in process".

The additional criterion *All messages* shows the messages of all users that are currently open or in process. You require the global permission for messages to view all messages for all users.

The application provides the following selection criteria:

Current

Event

You can select the event that caused messages.

History

Point in time ... to ...

You can select a period to restrict the data displayed.

Status

You can select a status. You can choose from the following options:

- in process
- open
- finished

Message status

You can select the message status. You can choose from the following options:

- read
- not read
- not processed
- forwarded

Event

You can select the event that caused messages.

Show messages of all users

If you enable this option, the overview does not only show your "own" escalations but that of all users.

You can select several escalations (multiple selection option) and use the functions *Forward* and *Complete messages* for the selected escalations at the same time.

Field descriptions

Reference

Unique ID of the message

Status

Message status

Priority

Priority of the message

Point in time

Date and time of the generation

Description

Stored description

Subject

Stored subject

Text

Stored text

Event

Triggered event

Message status

Message status

Comment

Stored comment

Status (recipient)

Displays the overall status of all notification functions. This status is "ERR" if a notification failed.

The following statuses exist:

NEW: new, not yet processed

ERR: error, not sent

OK: successfully sent

IP: in process

NON: no notification required

IMP: "impossible". The message cannot be sent (e.g. if the message is supposed to be sent by e-mail, but the person does not have an e-mail address).

Name

The recipient's first and last name.

Function

The recipient's function.

Person

The recipient's personnel number.

Triggered

Date / time and person who last edited the data record

Modified on

Date / time and person who last edited the data record

Finished

Date / time and person who last edited the data record

Forwarded

Date / time and person who last edited the data record

Toolbar**Read event**

View/read a new message/escalation. Once you have viewed a message, the reaction time is stopped and the processing time starts. The system saves the date and time.

Forward event

Forward a message/escalation to a new recipient.

Finish event

Complete a message/escalation.

5 Escalation history

Overview

HYDRA menu	Information management → Messages → Escalation history
FEDRA menu	Detailed scheduling → Master data → Escalation history
Transaction code	escev
Function authorization	escev



In the application Escalation history, the logged on user is not filtered in the displayed escalation. This means that all escalations meeting the selection criteria are displayed, even if they were originally intended for other recipients / users.

Depending on the escalations used, it may be necessary not to assign the function authorization for the escalation history.

Purpose

You use the escalation history if you want to get an overview of already completed escalations.

Field descriptions

Status

Escalation status

Event

This event triggers an escalation. Applications generate the event if specific situations occur.

Name

You can enter a description of the event in this field.

Subject

Title of the escalation message.

Text

Text of the escalation message.

Edited (date/ time / user)

Date and time stamp of the last editing.

Start (date/ time / user)

Date and time stamp when the escalation was started.

Last editing (date / time / user)

Date and time stamp of the last editing.

First editing (date / time / user)

Date and time stamp when the escalation was confirmed for the first time.

6 Escalation Management Archiving

Archiving

Escalation management events are archived after defined periods. The central archiving script hyarc.scr triggers the archiving process. This program is planned to be run on a daily basis within the [Scheduler](#) by default.

Subject to the given characteristics, archiving is performed either by the archiving program hyeskarc.exe/out or by [HYDRA Data Management](#).

How can I find out which archiving type is in use?

Check (you or your system administrator) whether or not the script hyarc.scr within the HYDRA directory on the HYDRA server includes the below entry:

```
# from hyeskarc.scr
# ESK deletion script (only if HYD-ESK)
if [ `hyliz.exe -r HYD-ESK` -gt 0 ]
then
    echo "HYD-ESK:" >> $ERRPATH/hyarc.pro
    hyeskarc.scr "ESK:D=7|ESK:A=M|A_ESK:D=30|A_ESK:A=X"
    cat $ERRPATH/hyeskarc.pro >> $ERRPATH/hyarc.pro
fi
```

If this is the case, archiving is still performed by the separate archiving program hyeskarc.exe/out.

If the entry does not exist or is commented out by inputting "#" in front of each line, archiving is performed based on [HYDRA Data Management](#).

Archiving by hyeskarc.exe/out

This program is started from the central archiving script hyarc.scr. This program is planned to be run on a daily basis within the [Scheduler](#) by default.

By default, archiving of the escalation management module is configured as follows:

Option	Value
Activity for data included in the online area (ESK:A=)	Archiving

Option	Value
Retention period within the online area (ESK:D=)	7 days
Activity for data included in the archive area (A_ESK:A)	Export to file
Retention period within the archive area (A_ESK:D)	30 days

Archiving using Data Management

In this case, configuration is made using the [HYDRA Data Management...\\functions\moc\MOC_DataManagement.pdf](#). Archiving is still started by the hyarc.scr script.

When transferring data into archive tables, such data is taken over the “retention period” of which (number in days/months/years; see values in brackets in the below table) has been exceeded.

Product	Object	Object designation	Transfer	Default interval
HYD-ESK	ESK	Escalations	Online stock → medium-term archive	7 days
HYD-ESK	A_ESK	Long-term archiving: Escalations	Medium-term archive → Long-term archive	30 days

7 Application-Relevant Settings in HYDRA

Maintenance of basic settings

You maintain the [basic settings](#) for Escalation Management in order to configure the following:

- SMTP Server
- SMTP Port
- SMTP Timeout
- SMTP Sender
- Other settings